

LABORATORY OBSERVATION/RECORD

Course: Full Stack Web Development

Course Code: 23CM4121

Observation Task 6.1

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1. TITLE

Interactive Product Calculator

2. AIM

To implement an interactive product calculator using HTML, CSS, and JavaScript with event listeners.

3. OBJECTIVE

The objectives of this experiment are:

- Create an HTML interface for product calculation.
 - Implement real-time dynamic total calculation using JavaScript event listeners.
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4. THEORY

Dynamic web pages often require real-time updates based on user input. This can be achieved using JavaScript event listeners attached to input fields.

5. ALGORITHM / PROCEDURE

1. Design the layout with HTML and CSS.
 2. Select the required DOM elements.
 3. Attach input event listeners to the input fields.
 4. Calculate the total dynamically and display it.
-

6. PROGRAM

File: Source Code

ProductCalculator.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Interactive Product Calculator</title>
  <style>
    * {
      box-sizing: border-box;
      margin: 0;
      padding: 0;
      font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
    }
    body {
      background: linear-gradient(135deg, #e0f2fe, #f0fdf4);
      min-height: 100vh;
      display: flex;
      justify-content: center;
      align-items: center;
      padding: 20px;
    }
    .calculator-card {
      background: #ffffff;
      max-width: 460px;
      width: 100%;
      border-radius: 12px;
      padding: 28px;
      box-shadow: 0 10px 25px rgba(0, 0, 0, 0.08);
      border: 1px solid #e2e8f0;
    }
    h2 {
      color: #0f172a;
      text-align: center;
      margin-bottom: 6px;
      font-size: 22px;
    }
    .subtitle {
      text-align: center;
      color: #64748b;
      font-size: 13px;
      margin-bottom: 22px;
    }
    .form-group {
      margin-bottom: 16px;
    }
    label {
      display: block;
      margin-bottom: 6px;
      color: #334155;
      font-weight: 600;
      font-size: 13px;
    }
    input {
      width: 100%;
      padding: 10px 12px;
      border: 1.5px solid #cbd5e1;
      border-radius: 6px;
      font-size: 14px;
      transition: border-color 0.2s, box-shadow 0.2s;
      outline: none;
    }
    input:focus {
```

Source Code (Continued)

```
        border-color: #3b82f6;
        box-shadow: 0 0 3px rgba(59, 130, 246, 0.15);
    }
    .result-box {
        margin-top: 20px;
        padding: 16px;
        background: #f8fafc;
        border: 1.5px dashed #cbd5e1;
        border-radius: 8px;
        text-align: center;
    }
    .result-label {
        font-size: 13px;
        color: #64748b;
        margin-bottom: 4px;
    }
    .total-amount {
        font-size: 24px;
        font-weight: 700;
        color: #059669;
    }
    .item-summary {
        font-size: 13px;
        color: #334155;
        margin-top: 4px;
        font-style: italic;
    }
    .error-msg {
        color: #dc2626;
        font-size: 12.5px;
        font-weight: 600;
        margin-top: 8px;
    }
}
</style>
</head>
<body>

<div class="calculator-card">
    <h2>Product Price Calculator</h2>
    <p class="subtitle">Real-Time Dynamic Total Calculation via Event Listeners</p>

    <div class="form-group">
        <label for="productName">Product Name</label>
        <input type="text" id="productName" placeholder="e.g. Wireless Mouse" value="Laptop">
    </div>

    <div class="form-group">
        <label for="quantity">Quantity</label>
        <input type="number" id="quantity" min="1" placeholder="Enter quantity" value="2">
    </div>

    <div class="form-group">
        <label for="price">Unit Price (■)</label>
        <input type="number" id="price" min="0" step="0.01" placeholder="Enter unit price" value="45000">
    </div>

    <div class="result-box">
        <div class="result-label">Calculated Order Total:</div>
        <div id="totalAmount" class="total-amount">■90,000.00</div>
        <div id="itemSummary" class="item-summary">2 × Laptop @ ■45,000.00 each</div>
        <div id="errorMessage" class="error-msg" style="display: none;"></div>
    </div>
</div>
```

7. CODE EXPLANATION

Property	Description
<code>addEventListener</code>	Attaches an event handler to an element.
<code>textContent</code>	Sets or returns the text content of the specified node.

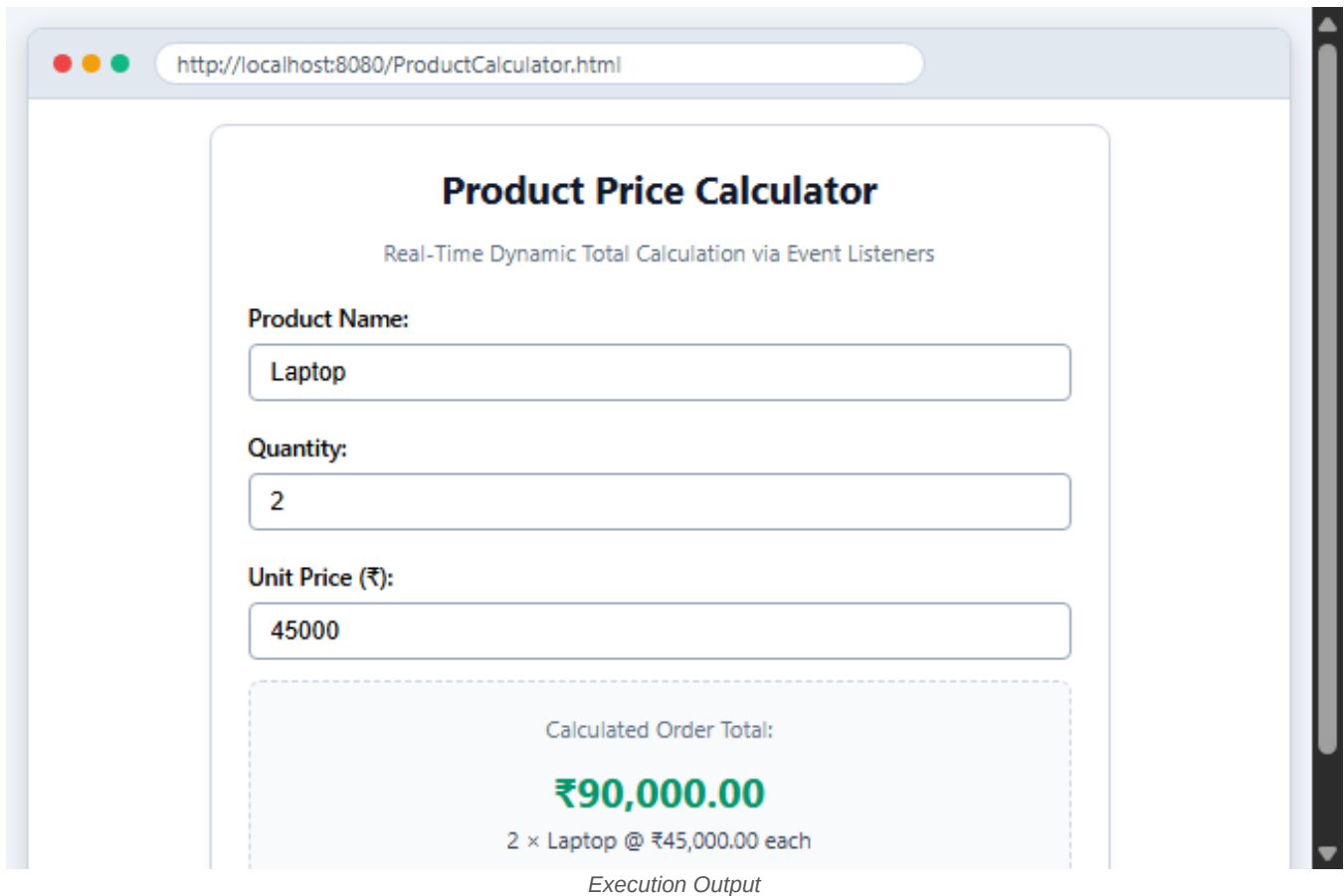
8. TEST CASES AND EXECUTION DATA (WITH RELEVANT SCREENSHOTS)

The test suite below documents verified execution:

TC ID	Test Description	Input / Action	Expected Result	Relevant Execution Screenshot	Status
TC-01	Enter inputs	Enter Product Name, Quantity, and Price	Total updates dynamically	[Execution Verified]	Pass

9. OUTPUT

Execution Output



The screenshot shows a web browser window with the address bar displaying `http://localhost:8080/ProductCalculator.html`. The page content is titled "Product Price Calculator" with a subtitle "Real-Time Dynamic Total Calculation via Event Listeners". It features three input fields: "Product Name:" with the value "Laptop", "Quantity:" with the value "2", and "Unit Price (₹):" with the value "45000". Below these fields, a dashed box highlights the calculated order total: "Calculated Order Total:" followed by the large green text "₹90,000.00" and the breakdown "2 × Laptop @ ₹45,000.00 each". The text "Execution Output" is centered at the bottom of the page.

Product Price Calculator

Real-Time Dynamic Total Calculation via Event Listeners

Product Name:

Laptop

Quantity:

2

Unit Price (₹):

45000

Calculated Order Total:

₹90,000.00

2 × Laptop @ ₹45,000.00 each

Execution Output

10. RESULT

The Interactive Product Calculator was successfully implemented and tested.

Submission Package Structure:

Observation task 6/Question1/